3.15 Factorisation

For this section we assume A is a commutative integral domain.

Definition 3.15.1 (Associates)
*We say two non-zero elements x and y are **associates** if $x = uy$ for some $u \in A^\star$. We write $x \sim y$.*

Note this is an equivalence relation on $A \setminus \{0\}$.

Lemma 3.15.2
Let A be a ring and $x, y \in A$ non-zero elements then the following are equivalent

- $x \mid y$
- $(y) \subseteq (x)$
- $y \in (x)$

Lemma 3.15.3
Let A be a ring and $x, y \in A$ non-zero elements then the following are equivalent

- $x \mid y$ *and* $y \mid x$
- $(x) = (y)$

If A is an integral domain this is equivalent to $x \sim y$.

Definition 3.15.4 (Irreducible element)
*We say $0 \neq x$ is **irreducible** if it is not invertible and $x = ab \implies a$ a unit or b a unit.*

Equivalently if $y \mid x$ implies either y is a unit or $y \sim x$.

Definition 3.15.5 (Prime element)
We say $0 \neq p$ is prime if $p \mid ab \implies p \mid a$ or $p \mid b$.

Example 3.15.6
The units of $\mathbb{Z}$ are $\{-1, 1\}$ so each equivalence class is of the form $\{n, -n\}$.

Example 3.15.7
A number $p \in \mathbb{Z}$ is prime in the traditional sense exactly when it is irreducible. It is of course also prime in the ring-theoretic sense but this requires proof (see (2.2.13)).

The concept of associates is important to unique factorization, because we may only hope to have unique factorization upto multiplication by a unit.

Lemma 3.15.8
If $x \sim y$ are associates then x is irreducible iff y is

Proof. Suppose $x \sim y$ and x irreducible. If $y = ab$ then $x = abu \implies a$ a unit or bu a unit $\implies b$ a unit. Therefore y is irreducible as required. ∎

Lemma 3.15.9 (Criterion for primality)
Suppose $0 \neq p \in \mathbb{Z}$. Then p is prime if and only if (p) is a prime ideal

Proof. Note $x \mid y \iff y \in (x)$. So in particular if p is prime then $xy \in (p) \implies p \mid xy \implies p \mid x$ or $p \mid y \implies x \in (p)$ or $y \in (p)$, whence (p) is prime.

Conversely if (p) is prime, then $p \mid xy \implies xy \in (p) \implies x \in (p)$ or $y \in p \implies p \mid x$ or $p \mid y$, so that p is prime. ∎

Lemma 3.15.10 (Criterion for irreducibility)
Let A be an integral domain. Then f is irreducible if and only if (f) is maximal amongst proper principal ideals.

Proof. Suppose f is irreducible and $(f) \subseteq (g)$. Then $f = ag$ with either a a unit or g a unit. If a is a unit then $(f) = (g)$, and if g is a unit $(g) = A$. So the result follows.

Conversely suppose $f = ab$, then $f \in (a) \implies (f) \subseteq (a)$. Then by hypothesis either $(a) = (f)$ or $(a) = A$. In the second case a is a unit. In the first case then $f \mid a \implies bf \mid f \implies b \mid 1$ whence b is a unit. ∎

Proposition 3.15.11 (Primes are Irreducible)
Let A be an integral domain then p prime $\implies p$ irreducible

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Proof. Suppose $b \mid p$ then $p = ab$ and $a \mid p$. By hypothesis $p \mid a$ or $p \mid b$. If $p \mid a$ (resp. b) then by (3.15.3) $p \sim a$ (resp. b) as required. ∎

Definition 3.15.12 (Unique Factorisation Domain (UFD) or Factorial Ring)
We say an integral domain A is factorial (or a UFD) if every element $0 \neq a$ may be represented as

$$a = u \prod_{i=1}^{n} p_i$$

for u a unit and p_i irreducible, and moreover this is unique in the sense that given another factorization

$$a = u' \prod_{i=1}^{m} p_i'$$

we have $n = m$ and $p_i \sim p'_{\psi(i)}$, for ψ a permutation on $\{1, ..., n\}$.

Furthermore it may be convenient in applications to count the multiplicities

Proposition 3.15.13 (Factorisation with multiplicities)
Let A be a UFD, then for every element $0 \neq a \in A$ there is a factorization of the form

$$a = u \prod_{i=1}^{n} p_i^{r_i}$$

where $r_i > 0$ and none of the p_i are associate to each other. Furthermore this is essentially unique in the sense that given another such factorization we have $n = n'$, $r_i = r'_{\sigma(i)}$ and $p_i \sim p'_{\sigma(i)}$ for some permutation $\sigma \in S_n$.

Proof. Given a factorization into irreducible elements

$$a = u \prod_{i=1}^{n} p_i$$

Consider a representative set of irreducibles $q_1, ..., q_m$ (under the equivalence relation $x \sim y$). Then we have $p_i = q_{\pi(i)} u_i$ for some units u_i and mapping $\pi : \{1, ..., n\} \to \{1, ..., m\}$. Let $r_j = \#\pi^{-1}(j)$. Then we have that the set of irreducibles $\{p_1, ..., p_n\}$ is the *disjoint* union of the set of equivalence classes with representatives q_j. Therefore

$$a = u \prod_{j=1}^{m} \prod_{p \sim q_j} p = u \prod_{j=1}^{m} \prod_{i:\pi(i)=j} u_i q_j = \left(u \prod_{j=1}^{m} \prod_{i:\pi(i)=j} u_i \right) \prod_{j=1}^{m} q_j^{r_j}$$

as required. Suppose we have two factorizations

$$u \prod_{i=1}^{n} p_i^{r_i} = u' \prod_{i=1}^{m} (p_i')^{r_i'}$$

Let I be the indexing set of p_i and J the set of p_j'. By unique factorization there must be mappings $\sigma : I \to J$ such that $p_i \sim p'_{\sigma(i)}$, and $\tau : J \to I$ such that $p_j' \sim p_{\tau(j)}$. Which means that $p_i \sim p_{\tau(\sigma(i))}$ and $p_j' \sim p'_{\sigma(\tau(j))}$. Since none are associate to each other we see that τ and σ are mutual inverses, whence $m = n$ and we may regard $\sigma \in S_n$. In the unique factorization p_i appears r_i times and $p'_{\sigma(i)}$ appears $r'_{\sigma(i)}$ times. Since p_i is associate to $p'_{\sigma(i)}$ it is not associate to any p_j' for $j \neq \sigma(i)$. Unique factorization shows that $r_i = r'_{\sigma(i)}$. ∎

Definition 3.15.14
Let A be a UFD and $x \in A$ a non-zero, non-unit such that

$$x \sim \prod_{i=1}^{n} p_i^{r_i}$$

is an (almost) unique factorization into irreducibles. Then for $p \in A$ an irreducible define

$$v_p(x) := \begin{cases} r_i & p \sim p_i \\ 0 & \text{otherwise} \end{cases}$$

If $x \in A$ is a unit then simply define $v_p(x) = 0$ for all p.

Lemma 3.15.15
Let A be a UFD then the following are equivalent

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a) $x \mid y$
b) $(y) \subseteq (x)$
c) $v_p(x) \leq v_p(y)$ *for all* p *irreducible*

In particular $x \sim y \iff (x) = (y) \iff v_p(x) = v_p(y)$ *for all* p *irreducible.*

Definition 3.15.16 (Atomic Ring)
We say that A *is* ***atomic*** *if every non-zero element has a (not necessarily unique) decomposition into irreducible elements (a unit being the empty product).*

Definition 3.15.17 (Ascending Chain Condition for Principal Ideals (ACCP))
We say a ring A *satisfies* ***ACCP*** *if every ascending chain of principal ideals eventually stabilizes.*

Note every Noetherian ring satisfies this condition.

Proposition 3.15.18
An integral domain A *satisfying ACCP is atomic.*

In particular a Noetherian ring is atomic.

Proof. Suppose a ring A is not atomic, then choose a non-zero non-unit $x_1 \in A$ with no decomposition into irreducibles. Then x_1 is not irreducible, so $x_1 = ab$ with a, b non-units, and one of a, b, say a, has no decomposition into irreducibles, as otherwise the product of the decompositions would decompose x_1. Put $x_2 := a$; then $(x_1) \subsetneq (x_2)$, for otherwise b would be a unit (3.15.10). Repeating this it's possible to construct a strictly ascending sequence of proper principal ideals

$$(x_1) \subsetneq (x_2) \subsetneq \ldots \subsetneq (x_n) \subsetneq \ldots$$

therefore A does not satisfy ACCP. ∎

Remark 3.15.19
The converse is not in general true [Gra74] but see (3.15.21) for a partial converse.

We show a simple criterion for a ring to be a UFD.

Definition 3.15.20
We say an integral domain A *is* ***AP*** *if* p *irreducible* $\implies$ p *prime.*

Roughly speaking, "atomic" ensures the existence of factorization and "AP" ensures the uniqueness.

Proposition 3.15.21 (Atomic + AP $\iff$ UFD)
Let A *be an integral domain. The following are equivalent*
a) A *is a UFD*
b) A *is atomic and AP*
c) A *satisfies ACCP and is AP*

Proof. *a)* $\implies$ *c)*. Suppose A is a UFD. Let p be an irreducible element and $p \mid ab$, say $ab = pc$. If $ab = 0$ then $a = 0$ or $b = 0$ and p divides it. Otherwise $a, b, c \neq 0$, and p times a factorization of c and the product of factorizations of a and b are two factorizations of ab, so by uniqueness p is associate to an irreducible occurring in the factorization of a or of b, whence $p \mid a$ or $p \mid b$. Therefore p is prime. Let $(x_1) \subseteq (x_2) \subseteq \ldots$ be an ascending chain of principal ideals. Discarding initial terms we may assume $x_1 \neq 0$, since $x_{i+1} \mid x_i$ and $x_{i+1} = 0$ force $x_i = 0$. Then by (3.15.15) $v_p(x_i)$ is a decreasing sequence in $\mathbb{N}$ for every irreducible p, and $v_p(x_i) = 0$ unless p is associate to one of the finitely many irreducibles occurring in the factorization of x_1. A decreasing sequence in $\mathbb{N}$ is eventually constant, so there is N such that $v_p(x_i) = v_p(x_N)$ for all $i \geq N$ and all such p. Therefore $v_p(x_i) = v_p(x_{i+1})$ for all irreducible p and all $i \geq N$, and by (3.15.15) $(x_i) = (x_{i+1})$, that is the chain stabilises.

c) $\implies$ *b)*. This is (3.15.18).

b) $\implies$ *a)*. We require to show that factorization into irreducibles is unique up to associates. Suppose

$$\prod_{i=1}^{n} p_i \sim \prod_{j=1}^{m} p'_j$$

By convention an empty product is 1 and by hypothesis all the elements are in fact prime. If $n = 0$, then since p'_j is irreducible, it is not a unit and hence $m = 0$, and likewise $n = 0$ if $m = 0$, as a prime dividing 1 would be a unit. Otherwise $n, m \geq 1$ and consider p_1, then $p_1 \mid$ RHS, so by definition of prime we must have $p_1 \mid p'_j$ for some j. Since

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p_j' is irreducible and p_1 is not a unit, we have $p_1 \sim p_j'$. Since A is integral we may cancel these two to obtain an equivalence of smaller degree and we may proceed by induction. ∎

Lemma 3.15.22 (PID is an AP-domain)
Let A be a PID and $a \in A$. Then the following are equivalent

a) a is prime
b) a is irreducible
c) (a) is maximal

Proof. *a)* $\implies$ *b)* This holds for an arbitrary integral domain (3.15.11).

b) $\iff$ *c)* as all ideals are principal this follows from (3.15.10)

c) $\implies$ *a)* By (3.3.60) (a) is prime, whence a is prime by (3.15.9) ∎

Proposition 3.15.23
A PID is Noetherian UFD.

Furthermore (f is irreducible $\iff$ f is prime), and every prime ideal is maximal.

Proof. A is Noetherian by (3.14.2) and atomic by (3.15.18). And by (3.15.22) an irreducible element is prime. Therefore we are done by (3.15.21). ∎

If we take a suitable fixed set of irreducible elements we can obtain completely unique factorization

Definition 3.15.24
*Let A be a ring we say $\mathcal{P}$ is a **representative set of irreducible elements** if*

- *No two elements $p, q \in \mathcal{P}$ are associate*
- *Every irreducible element $p \in A$ is associate to (precisely) one in $\mathcal{P}$*

Example 3.15.25
For $\mathbb{Z}$ the positive *primes are a canonical set of irreducible elements.*

Proposition 3.15.26
Let A be a UFD and $K = \operatorname{Frac}(A)$ then for every irreducible $p \in A$ there is a unique map

$$v_p : K^\star \to \mathbb{Z}$$

such that

- $u \in A^\star \implies v_p(u) = 0$
- $v_p(xy) = v_p(x) + v_p(y)$
- $v_p(p) = 1$

Furthermore let $\mathcal{P}$ be a set of irreducible representatives then we have a group isomorphism

$$\begin{aligned} K^\star/A^\star &\xrightarrow{\sim} \bigoplus_{p\in\mathcal{P}} \mathbb{Z} \\ x &\longrightarrow \left(v_p(x)\right)_{p\in\mathcal{P}} \\ \prod_{p\in\mathcal{P}} p^{n_p} &\longleftarrow \left(n_p\right)_{p\in\mathcal{P}} \end{aligned}$$

Finally $x \in A \iff v_p(x) \geq 0$ for all $p \in \mathcal{P}$.

Proof. By (3.15.13) there is a well-defined map $v_p : A \setminus \{0\} \to \mathbb{Z}$ satisfying the given properties. We claim that

$$\begin{aligned} v_p : K^\star &\to \mathbb{Z} \\ xy^{-1} &\to v_p(x) - v_p(y) \end{aligned}$$

is well-defined. For suppose $xy^{-1} = wz^{-1}$ then $zx = wy$ whence $v_p(z) + v_p(x) = v(w) + v(y)$ and therefore $v_p(xy^{-1}) = v_p(wz^{-1})$.

It's clear the multiplicative property also holds and so ϕ is well-defined (since by definition $A^\star$ is in the kernel of v_p). Denote by ϕ, ψ the proposed maps. By definition of unique factorization ϕ and ψ are mutual inverses when restricted as follows

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$$(A \setminus \{0\})/A^\star \leftrightarrow \bigoplus_{p \in \mathcal{P}} \mathbb{Z}_{\geq 0}$$

We also observe that $\phi(x^{-1}) = -\phi(x)$ and $\psi(-n) = \psi(n)^{-1}$, so then it's easy to demonstrate they are mutually inverse over the whole domain.

Suppose $v_p(xy^{-1}) \geq 0$ then $v_p(x) \geq v_p(y)$. If this holds for all $p \in \mathcal{P}$, then by (3.15.15) $y \mid x$ as elements of A whence by definition $xy^{-1} \in A$ as required. ∎

Proposition 3.15.27
Suppose A is an integral domain satisfying ACCP then so is $A[X]$.

Proof. Suppose we have an ascending chain of principal ideals

$$(f_1) \subseteq (f_2) \subseteq ...(f_n)...$$

Without loss of generality the f_i are non-zero. Then $f_{i+1} \mid f_i$ and by (...) $\deg(f_{i+1}) \leq \deg(f_i)$. Choose N such that $\deg(f_i)$ is minimal, and define $a_i := \ell(f_i) \in A$. Then by (3.7.3) $a_{i+1} \mid a_i$ for $i \geq N$ as elements of A. Therefore we have an increasing sequence of principal ideals

$$(a_N) \subseteq (a_{N+1}) \subseteq ...$$

which by hypothesis stabilizes, that is $a_i \sim a_j$ for all $i, j \geq M$ for some $M \geq N$. For $i \geq M$ we have $ua_i = a_{i+1}$, consider $uf_i - f_{i+1} \in (f_i)$. This has degree strictly smaller than N, and therefore by minimality must be 0. In particular $f_i \sim f_{i+1}$ and $(f_i) = (f_{i+1})$. ∎

Lemma 3.15.28
Suppose $p \in A$ is prime, then it is prime as an element of $A[X]$.

Lemma 3.15.29 (Nagata's Criterion)
Let A be a ring and S a multiplicative subset generated by prime elements and units. Let $f \in A$ be irreducible or a unit, then

a) $\frac{f}{1}$ is irreducible or a unit in $S^{-1}A$
b) $\frac{f}{1}$ prime or a unit in $S^{-1}A \implies f$ is a prime or a unit in A.

Furthermore if $S^{-1}A$ is AP then so is A.

Proof. Note the condition on S means every $a \in S$ satisfies $a \sim p_1...p_r$ for primes $p_i \in A$.

a) Suppose $\frac{f}{1} = \frac{g}{a}\frac{h}{b}$ for $f, g, h \in A$ and $a, b \in S$, then $abf = gh$. Further $ab \sim p_1...p_r$. Then $p_i \mid a$ or $p_i \mid b$, whence we can find $f \sim g'h'$ where $g = cg'$, $h = dh'$ and $c, d \in S$. As f is irreducible (or a unit), then for example g' is invertible, in which case $\frac{g}{a} = \frac{cg'}{a}$ is invertible. Therefore $\frac{f}{1}$ is either irreducible or a unit.
b) The case f a unit is clear, so assume that f is irreducible. Suppose $\frac{f}{1}$ is prime or a unit and $f \mid gh$. Then $\frac{f}{1} \mid \frac{g}{1}\frac{h}{1}$ and for example $\frac{f}{1} \mid \frac{g}{1}$. Therefore $ff' = gp_1...p_r$ for some $f' \in A$ and $p_1, ..., p_r \in A$ prime. If $p_i \mid f$ for some i then by irreducibility we have $p_i \sim f$, and we see that f is prime. Otherwise $p_i \mid f'$ for all i and we find $f \mid g$. Therefore f is prime as required.

The last statement follows immediately from the previous two results. ∎

3.15.1 Polynomials over a Field

In this section we consider the polynomial ring with coefficients in a field, $k[X]$.

Proposition 3.15.30 (Associate polynomials)
The following are equivalent for $0 \neq f, g$

- $f \sim g$
- $f = \lambda g$ *for* $\lambda \neq 0$
- $f \mid g$ *and* $g \mid f$

Proposition 3.15.31
A polynomial $f \in k[X]$ is associate to precisely one monic polynomial g. If f is irreducible so is g.

Proof. **TODO** ∎

Proposition 3.15.32 (Polynomial ring is a PID)
Let k be a field, then $k[X]$ is a PID, and therefore a Noetherian UFD.

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Proof. Let $(0) \neq \mathfrak{a}$ be an ideal and let $f \in \mathfrak{a}$ be a polynomial of minimal degree. We may assume it is monic. Any $g \in \mathfrak{a}$ may be represented as $f = qg + r$ by the division algorithm. Clearly $r \in \mathfrak{a}$, therefore by minimality $r = 0$, whence $g \in (f)$. ∎

Proposition 3.15.33 (Unique Factorisation of Polynomials)
For the ring $k[X]$ the set of irreducible monic polynomials constitutes a representative set (3.15.24). Therefore we have a unique factorization

$$f = \ell(f) \prod_{p \text{ irreducible monic}} p^{v_p(f)}$$

such that

$$v_p(fg) = v_p(f) + v_p(g)$$

Proof. (3.15.31) shows that the irreducible monic polynomials constitute a representative set. Therefore the result follows from (3.15.26). Let u be the unit appearing in the factorization, it must be an element of k. Compare leading coefficients to see that $u = \ell(f)$. ∎

Lemma 3.15.34 (Roots and Multiplicity)
For $f \in k[X]$ a non-constant polynomial and $\alpha \in k$ we have

$$f(\alpha) = 0 \iff (X - \alpha) \mid f \iff v_{X-\alpha}(f) > 0$$

In this case $r := v_{X-\alpha}(f)$ is the multiplicity of the root α, and observe

$$f(X) = \ell(f)(X - \alpha)^r g(X)$$

with $g(\alpha) \neq 0$ (equivalently $v_{X-\alpha}(g) = 0$).

Proof. The right to left implication is obvious. Conversely by the division algorithm we may write

$$f(X) = f(\alpha) + (X - \alpha)Q(X)$$

Then if $f(\alpha) = 0$ we clearly have $v_{X-\alpha}(f) > 0$. Finally we may construct

$$g(X) = \prod_{p \neq (X-\alpha)} p^{v_p(f)}$$

It's clear that for every p appearing in the product $p(\alpha) \neq 0$ because otherwise we would have $(X - \alpha) \mid p$ and by irreducibility $(X - \alpha) = p$. Therefore $g(\alpha) \neq 0$ as required. ∎

Corollary 3.15.35
Let K/k be a field extension and $0 \neq f \in k[X]$. Then f has at most $\deg(f)$ roots in K. In particular a polynomial of degree $< n$ with n distinct roots in K is zero.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Identify f with its image in $K[X]$, which has the same degree. Let $\alpha_1, ..., \alpha_m$ be the distinct roots of f in K and $r_i := v_{X-\alpha_i}(f) \geq 1$ their multiplicities in $K[X]$ (3.15.34). Then $f = \ell(f) \prod_{i=1}^{m} (X - \alpha_i)^{r_i} g$ where g is the product of the remaining irreducible factors of f in $K[X]$, which is a constant when f splits completely in K (3.17.8). Therefore $\deg(f) = \sum_i r_i + \deg(g) \geq m$. ∎

Lemma 3.15.36 (Lagrange Interpolation)
Let k be a field, $y_1, ..., y_n \in k$ distinct and $h \in k[X]$ with $\deg(h) < n$. Then

$$h(X) = \sum_{l=1}^{n} h(y_l) \prod_{k \neq l} \frac{X - y_k}{y_l - y_k}.$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

The difference of the two sides has degree $< n$ and vanishes at each y_j, as the l-th term of the sum vanishes at y_j for $l \neq j$ and equals $h(y_j)$ at y_j for $l = j$. It has n roots so is zero by (3.15.35). ∎

Proposition 3.15.37 (Criteria for Multiple Roots)
Let $f(X) \in k[X]$ be a polynomial, r a positive integer and suppose that either $\mathrm{char}(k) = 0$ or $r < \mathrm{char}(k)$. Then $\alpha \in k$ is a root of f with multiplicity r precisely when

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$$f(\alpha) = f^{(1)}(\alpha) = ... = f^{(r-1)}(\alpha) = 0$$

and $f^{(r)}(\alpha) \neq 0$.

Therefore the multiple roots are precisely the common roots of $f(X)$ *and* $f'(X)$ *(irrespective of the characteristic).*

Proof. Note that by (3.15.34) and (3.7.10)

$$f^{(1)}(X) = (X-\alpha)^{r-1}[rg(X) + (X-\alpha)g'(X)]$$

with $g(\alpha) \neq 0$ and r the multiplicity of the root. If $r = 1$, then $f^{(1)}(\alpha) = g(\alpha) \neq 0$ as required. If $r > 1$, then $f^{(1)}(X)$ has α as a root of multiplicity $r-1$, so it follows by induction.

The second statement is simply the case $r = 1$. ∎

Definition 3.15.38 (Separable Polynomial)
A polynomial $f \in k[X]$ *is* ***separable*** *if* f *and* f' *are co-maximal, that is* $(f, f') = (1)$. *Otherwise it is* ***inseparable****.*

Proposition 3.15.39
A separable polynomial $f \in k[X]$ *has no multiple roots (and* f *and* f' *have no common roots) in any extension field* K/k.

Proof. Since $(f, f') = (1)$ we have $af + bf' = 1$ for some $a, b \in k[X]$. Clearly f and f' can have no common roots, and therefore f has no multiple roots by (3.15.37). ∎

Proposition 3.15.40
Suppose $f, g \in k[X]$, g *is separable and* $f \mid g$, *then* f *is separable.*

Proof. Suppose f is not separable then by (3.14.5) f and f' have a common divisor d such that $\deg(d) > 0$. Since $g = fh$, so $g' = f'h + fh'$. Therefore d is also a non-trivial common divisor of g and g' contradicting (3.14.5). ∎

3.15.2 Polynomial Ring is a UFD

We prove the following result, first by Nagata's Criterion and again by Gauss' Lemma.

Proposition 3.15.41
Suppose A *is a UFD, then so is* $A[X]$.

Proof. By (3.15.21) we need to show that $A[X]$ satisfies ACCP and is AP. The first follows from (3.15.27).

Let $S = A \setminus \{0\}$ the set of non-zero elements. Let $K = \mathrm{Frac}(A)$. If we regard $A[X]$ as a subring of $K[X]$ then we claim $S^{-1}(A[X]) = K[X]$; this follows by multiplying an element of $K[X]$ by the product of denominators of all the coefficients. Furthermore as A is a UFD (and by (3.15.21)) S is generated by prime elements and units. As K is a field, $K[X]$ is a PID (3.15.32), so by (3.15.23) it is a UFD, and so in particular is AP. Therefore by (3.15.29) $A[X]$ is AP as required. ∎

For Gauss' Lemma we require to introduce some notation first.

Definition 3.15.42 (Primitive polynomials)
Let A *be a UFD,* $K := \mathrm{Frac}(A)$ *and* $f \in K[X]$ *a polynomial given by*

$$f(X) = \sum_{i=0}^{n} a_i X^i$$

Define the ***content*** *of* f *by*

$$c(f) = \prod_{p \in \mathcal{P}} p^{\min_i v_p(a_i)} \in K^\star$$

where the product is taken over a representative set of primes for A. *Changing the set of representatives only changes the value of* $c(f)$ *by multiplication of a unit.*

We say that f *is* ***primitive*** *precisely when* $c(f) = 1$

Lemma 3.15.43 (Gauss' Lemma I)
Let A *be a UFD and* $f \in K[X]$ *where* $K = \mathrm{Frac}(A)$. *Then the content* $c(f)$ *satisfies the following properties*

a) $f \in A[X] \iff c(f) \in A$
b) f *is primitive* $\iff \min_i v_p(a_i) = 0 \quad \forall p$ prime *and in this case* $f \in A[X]$
c) $c(\lambda f) = \lambda c(f)$ *for* $\lambda \in K^\star$ *up to multiplication by a unit in* A

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d) $f/c(f) \in A[X]$ *is primitive*
e) f, g *primitive* $\implies fg$ *primitive*
f) $c(fg) = c(f)c(g)$ *for all* $f, g \in K[X]$

Proof. We prove each in turn

a) $f \in A[X] \iff a_i \in A \quad \forall i \overset{(3.15.26)}{\iff} v_p(a_i) \geq 0 \; \forall i \; \forall p \in \mathcal{P} \iff \min_i v_p(a_i) \geq 0 \; \forall p \in \mathcal{P} \overset{(3.15.26)}{\iff} c(f) \in A$
b) $c(f) = 1 \iff v_p(c(f)) = 0 \quad \forall p \iff \min_i v_p(a_i) = 0 \quad \forall p$. Further $v_p(a_i) \geq 0$ whence $f \in A[X]$ by (3.15.26).
c) Note $v_p(\lambda a_i) = v_p(\lambda) + v_p(a_i) \implies v_p(c(\lambda f)) = \min_i v_p(\lambda a_i) = v_p(\lambda) + \min_i v_p(a_i) = v_p(\lambda) + v_p(c(f))$. Whence by (3.15.26) we see $c(\lambda f)$ and $\lambda c(f)$ are equal up to multiplication by a unit in A.
d) Clear by b).
e) Consider p prime and the reduction $\bar{\cdot} : A[X] \to (A/(p))[X]$. Then by assumption $\overline{f}$ and $\overline{g}$ are non-zero. Furthermore $(A/(p))[X]$ is an integral domain so that $\overline{f \cdot g} = \overline{f} \cdot \overline{g} \neq 0$. Therefore p does not divide all the coefficients of $f \cdot g$. As p was arbitrary this shows that $f \cdot g$ is primitive.
f) We may reduce to e) by dividing by the content.

■

Lemma 3.15.44 (Gauss' Lemma II)
Let A *be a UFD and* $f \in A[X]$*. Then the following are equivalent*

a) f *is irreducible in* $A[X]$
b) f *is an irreducible element of* A *or (*f *is primitive and irreducible in* $K[X]$*)*

where $K = \mathrm{Frac}(A)$.

In particular if $f \in K[X]$ *is irreducible then* $f/c(f)$ *is irreducible in* $A[X]$.

Proof. $b) \implies a)$. It's clear that an irreducible element of A remains irreducible in $A[X]$. Suppose $0 \neq f \in A[X]$ is primitive and irreducible in $K[X]$. Then as $f \notin K[X]^\star$ we have $\deg(f) > 0$. Suppose $f = gh$ in $A[X]$ then by irreducibility $\deg(g) = 0$ or $\deg(h) = 0$. Further $1 = c(f) = c(g)c(h)$ by Gauss' Lemma, so one of h and g must lie in $A^\star = A[X]^\star$. Therefore f is irreducible in $A[X]$ as required.

$a) \implies b)$. If $\deg(f) = 0$ then clearly f is an irreducible element of A. So assume $\deg(f) > 0$. Observe $f = c(f) \cdot (f/c(f))$ so by irreducibility we require $c(f) = 1$ and therefore f is primitive. Suppose $f = gh$ in $K[X]$ then $1 = c(g)c(h)$ and therefore

$$f = \frac{g}{c(g)} \frac{h}{c(h)}$$

is a decomposition in $A[X]$ which shows one of g, h has degree 0. Therefore f is irreducible in $K[X]$ as required. ■

Proposition 3.15.45
Let A *be a UFD. Then so is* $A[X]$.

Proof. For $0 \neq f \in A[X]$ we may use irreducible factorisation in $K[X]$ to find

$$f = \lambda \pi_1 ... \pi_n$$

where $\lambda \in K$ and $\pi_i \in K[X]$ are irreducible. Replace $\pi_i \to \pi_i / c(\pi_i)$ then may assume that π_i are irreducible in $A[X]$ by (3.15.44). Furthermore

$$c(f) = \lambda c(\pi_i) ... c(\pi_n) = \lambda$$

which shows $\lambda \in A$. As A is a UFD then λ may be decomposed into irreducibles of A, which by (3.15.44) are irreducible in $A[X]$. Therefore $A[X]$ is atomic.

Suppose that $f \in A[X]$ is irreducible we require to show it is prime. The case $\deg(f) = 0$ follows from the AP property of A, so assume $\deg(f) > 0$. Suppose $f \mid gh$ then as $K[X]$ is a UFD we see that, without loss of generality, $qf = g$ for $q \in K[X]$. Then $c(qf) = c(q)c(f) = c(q) = c(g) \in A$, so $q \in A[X]$ by (3.15.43) and we see $f \mid g$ in $A[X]$. This shows that $A[X]$ is AP and therefore a UFD by (3.15.21). ■

Corollary 3.15.46
Suppose A *is a UFD. Then* $A[X_1, ..., X_n]$ *is a UFD.*

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